



Binomial Theorem: RACE-01

Single Correct Type Questions:

1. 16^{th} term in the expansion of $(\sqrt{x} - \sqrt{y})^{17}$ is
 (A) $136xy^7$ (B) $136xy$ (C) $-136xy^{15/2}$ (D) $-136xy^2$ MBT167
2. Coefficient of x in the expansion of $\left(x^2 + \frac{a}{x}\right)^5$ is
 (A) $9a^2$ (B) $10a^3$ (C) $10a^2$ (D) $10a$ MBT168
3. The coefficient of x^7 in the expansion of $\left(\frac{x^2}{2} - \frac{2}{x}\right)^8$ is
 (A) -56 (B) 56 (C) -14 (D) 14 MBT169
4. In the expansion of $\left(x - \frac{1}{x}\right)^6$, the term independent of x is
 (A) -20 (B) 20 (C) 30 (D) -30 MBT170
5. The coefficient of x^{32} in the expansion of $\left(x^4 - \frac{1}{x^3}\right)^{15}$ is
 (A) ${}^{15}C_5$ (B) ${}^{15}C_6$ (C) ${}^{15}C_4$ (D) ${}^{15}C_7$ MBT171
6. The term independent of x in the expansion of $\left(\frac{1}{2}x^{1/3} + x^{-1/5}\right)^8$ will be
 (A) 5 (B) 6 (C) 7 (D) 8 MBT172
7. In the expansion of $(1 + x + x^3 + x^4)^{10}$ the coefficient of x^4 is
 (A) ${}^{40}C_4$ (B) ${}^{10}C_4$ (C) 210 (D) 310 MBT173
8. The coefficient of middle term in the expansion of $(1 + x)^{10}$ is
 (A) $\frac{10!}{5!6!}$ (B) $\frac{10!}{(5!)^2}$ (C) $\frac{10!}{5!7!}$ (D) None of these MBT174
9. The number of dissimilar terms in the expansion of $(1 + x^4 - 2x^2)^{15}$ is
 (A) 21 (B) 31 (C) 41 (D) 61 MBT175

10. Find the middle term(s) in the expansion of $\left(2y - \frac{y^2}{2}\right)^{11}$

(A) $-2^{11}C_5y^{16} & \frac{^{11}C_6}{2}y^{17}$

(B) $2^{11}C_5y^{16} & \frac{^{11}C_6}{2}y^{17}$

(C) $2^{11}C_5y^{16} & -\frac{^{11}C_6}{2}y^{17}$

(D) $-2^{11}C_5y^{16} & -\frac{^{11}C_6}{2}y^{17}$

MBT176

11. The value of $\frac{18^3+7^3+3.18.7.25}{3^6+6.243.2+15.81.4+20.27.8+15.9.16+6.3.32+64}$ is :

(A) 1

(B) 2

(C) 3

(D) None

MBT177

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	C	B	C	A	C	C	D	B	B	A	A



Binomial Theorem: RACE-02

Single Correct Type Questions:

1. ${}^{15}C_{3r} = {}^{15}C_{r+3}$ then value of r is
 (A) 3 (B) 4
 (C) 5 (D) 8
 MBT178
2. The value of ${}^{15}C_3 + {}^{15}C_{13}$ is
 (A) ${}^{16}C_3$ (B) ${}^{30}C_{16}$
 (C) ${}^{15}C_{10}$ (D) ${}^{16}C_{14}$
 MBT179
3. If $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + \dots + {}^nC_nx^n$, then the value of ${}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n$ is -
 (A) 2^{n+1} (B) 2^{n-1}
 (C) $2^n + 1$ (D) $2^n - 1$
 MBT180
4. If $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n$ then the value of ${}^nC_0 + 3 \cdot {}^nC_1 + 5 \cdot {}^nC_2 + \dots + (2n+1) \cdot {}^nC_n$ is -
 (A) $n \cdot 2^n$ (B) $(n-1) \cdot 2^n$
 (C) $(n+2) \cdot 2^{n-1}$ (D) $(n+1) \cdot 2^n$
 MBT181
5. The value of $\sum_{r=1}^{10} r \cdot \frac{{}^nC_r}{{}^nC_{r-1}}$ is equal to :
 (A) $5(2n-9)$ (B) $10n$
 (C) $9(n-4)$ (D) $n-4$
 MBT182
6. If $(1+x)^n = \sum_{r=0}^n a_r x^r$, $b_r = 1 + \frac{a_r}{a_r - 1}$ and $\prod_{r=1}^n b_r = \frac{(101)^{100}}{100!}$, then n equals to :
 (A) 99 (B) 100
 (C) 101 (D) 102
 MBT183
7. $\sum_{r=0}^{n-1} \frac{{}^nC_r}{{}^nC_r + {}^nC_{r+1}} =$
 (A) $\frac{n}{2}$ (B) $\frac{n+1}{2}$
 (C) $(n+1) \frac{n}{2}$ (D) $\frac{n(n-1)}{2(n+1)}$
 MBT184

8. $\frac{{}^{11}C_0}{1} + \frac{{}^{11}C_1}{2} + \frac{{}^{11}C_2}{3} + \dots + \frac{{}^{11}C_{10}}{11} =$

(A) $\frac{2^{11}-1}{11}$

(B) $\frac{2^{11}-1}{6}$

(C) $\frac{3^{11}-1}{11}$

(D) $\frac{3^{11}-1}{6}$

MBT185

9. If the sum of the coefficients in the expansion of $(2 + 3cx + c^2x^2)^{12}$ vanishes, then c equals to

(A) -1, 2

(B) 1, 2

(C) 1, -2

(D) -1, -2

MBT186

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	A	A	D	D	A	B	A	B	D



Binomial Theorem: RACE-03

Single Correct Type Questions:

1. The remainder when 17^{10} is divided by 9 is:
 (A) 8 (B) 7 (C) 2 (D) 1 MBT187
2. The remainder when 7^{98} is divided by 5 is
 (A) 4 (B) 0 (C) 2 (D) 3 MBT188
3. The remainder when 2^{2003} is divided by 17 is :
 (A) 1 (B) 2 (C) 8 (D) none of these MBT189
4. The last three digits of 3^{50} will be
 (A) 249 (B) 259 (C) 349 (D) 241 MBT190
5. The last two digits of the number 3^{400} is :
 (A) 81 (B) 43 (C) 29 (D) 01 MBT192
6. If $n \in N$ then $10^n + 5$ is divisible by :
 (A) 3 (B) 7 (C) 9 (D) 2 MBT193
7. If the numerically greatest term in the expansion of $(3-5x)^{15}$ when $x = \frac{1}{5}$ is 455×3^n , $n \in N$ then value of n is
 (A) 10 (B) 12 (C) 14 (D) 16 MBT194
8. The greatest term in the expansion of $\sqrt{3} \left(1 + \frac{1}{\sqrt{3}}\right)^{20}$ is
 (A) $\frac{25840}{9}$ (B) $\frac{24840}{9}$ (C) $\frac{26840}{9}$ (D) None of these MBT195

ANSWER KEY

Question	1	2	3	4	5	6	7	8	
Answer	D	A	C	A	D	A	B	A	



Binomial Theorem: RACE-04

Single Correct Type Questions:

1. The sum of the coefficients of even powers of x in the expansion of $(1 + x + x^2 + x^3)^5$ is -
 (A) 512 (B) -512 (C) 215 (D) none of these

MBT196

2. The value of the expression $\left(\sum_{r=0}^{10} {}^{10}C_r\right)\left(\sum_{K=0}^{10} (-1)^K \frac{{}^{10}C_K}{2^K}\right)$ is :
 (A) 2^{10} (B) 2^{20} (C) 1 (D) 2^5

MBT197

3. The value of $\binom{50}{0}\binom{50}{1} + \binom{50}{1}\binom{50}{2} + \dots + \binom{50}{49}\binom{50}{50}$ is, where ${}^nC_r = \binom{n}{r}$
 (A) $\binom{100}{50}$ (B) $\binom{100}{51}$ (C) $\binom{50}{25}$ (D) $\binom{50}{25}^2$

MBT198

4. In the expansion of $(1+x)^n \left(1+\frac{1}{x}\right)^n$, the term independent of x is -
 (A) $C_0^2 + 2C_1^2 + \dots + (n+1)C_n^2$ (B) $(C_0 + C_1 + \dots + C_n)^2$
 (C) $C_0^2 + C_1^2 + \dots + C_n^2$ (D) None of these

MBT199

5. If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$, then find the value of $C_0^2 + C_1^2 + C_2^2 + \dots + C_n^2 =$
 (A) nC_n (B) $2^n C_n$ (C) $3^n C_n$ (D) $4^n C_n$

MBT200

6. If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$, then find the value of $C_0C_2 + C_1C_3 + C_2C_4 + \dots + C_{n-2}C_n =$
 (A) $2^n C_{n-3}$ (B) ${}^nC_{n-2}$ (C) $2^n C_{n-2}$ (D) $2^n C_{n-4}$

MBT201

7. If $|x| < 1$ then 7th term in the expansion of $(1-x)^{-2}$ is
 (A) $7x^7$ (B) $6x^7$ (C) $7x^6$ (D) $6x^6$

MBT202

8. $\left(x^2 + \frac{1}{x}\right)^{-4/3}$ can be expanded by binomial theorem if
 (A) $x < 1$ (B) $|x| < 1$ (C) $x > 1$ (D) $|x| > 1$

MBT203

9. If $|x| < 1$, then the co-efficient of x^n in the expansion of $(1 + x + x^2 + x^3 + \dots)^2$ is
 (A) n (B) $n-1$ (C) $n+2$ (D) $n+1$

MBT204

10. The coefficient of x^4 in the expression $(1 + 2x + 3x^2 + 4x^3 + \dots \text{upto } \infty)^{\frac{1}{2}}$ (where $|x| < 1$) is
 (A) 1 (B) 3 (C) 2 (D) 5

MBT205

Multiple Correct Type Questions:

11. The coefficient of x^4 in $\left(\frac{1+x}{1-x}\right)^2$, $|x| < 1$, is
 (A) 4 (B) -4 (C) $10 + {}^4C_2$ (D) 16
12. The expansion of $(3x + 2)^{-1/2}$ is valid in ascending powers of x , if x lies in the interval.
 (A) $(0, 2/3)$ (B) $(-3/2, 3/2)$ (C) $(-2/3, 2/3)$ (D) $(-\infty, -3/2) \cup (3/2, \infty)$

MBT206

MBT207

13. If $(3\sqrt{3} + 5)^n = p + f$, where p is an integer and f is a proper fraction, then find the value of $(3\sqrt{3} - 5)^n$, $n \in N$, is
 (A) $1 - f$, if n is even (B) f , if n is even (C) $1 - f$, if n is odd (D) f , if n is odd

MBT208

14. If $(9 + \sqrt{80})^n = I + f$, where I, n are integers and $0 < f < 1$, then:
 (A) I is an odd integer (B) I is an even integer
 (C) $(I + f)(1 - f) = 1$ (D) $1 - f = (9 - \sqrt{80})^n$

MBT209

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	B	C	B	C	C	D	D	A
Question	11	12	13	14						
Answer	C,D	A,C	A,D	A,C,D						



Binomial Theorem: RACE-05

Single Correct Type Questions:

1. The value of $\binom{30}{0} \binom{30}{10} - \binom{30}{1} \binom{30}{11} + \binom{30}{2} \binom{30}{12} \dots + \binom{30}{20} \binom{30}{30}$, is where $\binom{n}{r} = {}^nC_r$
- (A) ${}^{30}C_{10}$ (B) ${}^{60}C_{20}$
(C) ${}^{31}C_{11}$ or ${}^{31}C_{10}$ (D) ${}^{30}C_{11}$

MBT210

2. For $r = 0, 1, \dots, 10$, let A_r, B_r and C_r denote, respectively, the coefficient of x^r in the expansions of $(1+x)^{10}, (1+x)^{20}$ and $(1+x)^{30}$. Then $\sum_{r=1}^{10} A_r(B_{10}B_r - C_{10}A_r)$ is equal to :
- (A) $B_{10} - C_{10}$ (B) $A_{10}(B_{10}^2 - C_{10}A_{10})$
(C) 0 (D) $C_{10} - B_{10}$

MBT211

3. Coefficient of x^{11} in the expansion of $(1+x^2)^4(1+x^3)^7(1+x^4)^{12}$ is -
- (A) 1051 (B) 1106
(C) 1113 (D) 1120

MBT212

4. The sum of coefficients of integral powers of x in the binomial expansion of $(1 - 2\sqrt{x})^{50}$ is :
- (A) $\frac{1}{2}(3^{50} - 1)$ (B) $\frac{1}{2}(2^{50} + 1)$
(C) $\frac{1}{2}(3^{50} + 1)$ (D) $\frac{1}{2}(3^{50})$

MBT213

5. Statement -1 : $\sum_{r=0}^n (r+1) {}^nC_r = (n+2)2^{n-1}$

Statement - 2 : $\sum_{r=0}^n (r+1) {}^nC_r x^r = (1+x)^n + nx(1+x)^{n-1}$

- (A) Statement -1 is false, Statement -2 is true
(B) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1
(C) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1
(D) Statement-1 is true, Statement-2 is false

MBT214

6. The remainder left out when $8^{2^n} - (62)^{2^{n+1}}$ is divided by 9 is :
- (A) 7 (B) 8
(C) 0 (D) 2

MBT215

7. Let $S_1 = \sum_{j=1}^{10} j(j-1) {}^{10}C_j$, $S_2 = \sum_{j=1}^{10} j {}^{10}C_j$ and $S_3 = \sum_{j=1}^{10} j^2 {}^{10}C_j$

Statement-1 : $S_3 = 55 \times 2^9$.

Statement-2 : $S_1 = 90 \times 2^8$ and $S_2 = 10 \times 2^8$.

- (A) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for statement-1.
 (C) Statement-1 is true, Statement-2 is false.
 (D) Statement-1 is false, Statement-2 is true.

MBT216

8. The coefficient of x^7 in the expansion of $(1-x-x^2+x^3)^6$ is :

- (A) -144 (B) 132 (C) 144 (D) -132

MBT217

9. If n is a positive integer, then $(\sqrt{3}+1)^{2n} - (\sqrt{3}-1)^{2n}$ is :

- (A) a rational number other than positive integers
 (B) an irrational number
 (C) an odd positive integer
 (D) an even positive integer

MBT218

10. The coefficients of three consecutive terms of $(1+x)^{n+5}$ are in the ratio 5: 10: 14. Then $n =$

- (A) 5 (B) 6 (C) 7 (D) 8

MBT219

11. $\sum_{r=0}^n (-1)^r {}^nC_r \cdot \frac{(1+r\ell n10)}{(1+\ell n10^n)^r} =$

- (A) 0 (B) $\frac{1}{2}$ (C) 1 (D) None of these

MBT220

12. Coefficient of x^{n-1} in the expansion of,

$$(x+3)^n + (x+3)^{n-1}(x+2) + (x+3)^{n-2}(x+2)^2 + \dots + (x+2)^n \text{ is :}$$

- (A) ${}^{n+1}C_2(3)$ (B) ${}^{n-1}C_2(5)$
 (C) ${}^{n+1}C_2(5)$ (D) ${}^nC_2(5)$

MBT221

13. If $(1+x)^n = \sum_{r=0}^n a_r x^r$ and $b_r = 1 + \frac{a_r}{a_{r-1}}$ and $\prod_{r=1}^n b_r = \frac{(101)^{100}}{100!}$, then n equals to :

- (A) 99 (B) 100 (C) 101 (D) 102

MBT222

14. Number of rational terms in the expansion of $(1 + \sqrt{2} + \sqrt{5})^6$ is :

- (A) 7 (B) 10 (C) 6 (D) 8

MBT223

15. If $S = {}^{404}C_4 - {}^{303}C_4 + {}^{202}C_4 - {}^{101}C_4 = (101)^k$ then k equals to :

- (A) 1 (B) 2 (C) 4 (D) 6

MBT224

Multiple Correct type Questions:

16. In the expansion of $(x + y + z)^{25}$
 (A) every term is of the form ${}^{25}C_r \cdot {}^rC_k \cdot x^{25-r} \cdot y^{r-k} \cdot z^k$
 (B) the coefficient of $x^8y^9z^9$ is 0
 (C) the number of terms is 325
 (D) none of these

MBT225

17. If $(1 + x + 2x^2)^{20} = a_0 + a_1x + a_2x^2 + \dots + a_{40}x^{40}$, then $a_0 + a_2 + a_4 + \dots + a_{38}$ is equal to
 (A) $2^{19}(2^{30} + 1)$
 (B) $2^{19}(2^{20} - 1)$
 (C) $2^{39} - 2^{19}$
 (D) $2^{39} + 2^{19}$

MBT226

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	D	C	C	B	C	C	A	B	B
Question	11	12	13	14	15	16	17			
Answer	A	C	B	B	C	A,B	B,C			